Practice Questions
DSC 102 Spring 2026

Short Questions

1. Minimum bits required to represent 70,000 unique items?
Ans: 17
Sol: (2^{16}<70000<2^{17})

2. Compute decimal value of (1011_2 + 1A_{16} - 12_8).
Ans: 27
Sol: (11 + 26 - 10)

3. 32-bit byte-addressable system with 8-byte aligned instructions; fraction of valid instruction addresses?
Ans: 1/8
Sol: Only multiples of 8 valid

4. CPU 2.5 GHz, 2B instructions; 40% ALU(1), 20% ctrl(2), 40% mem(21 cycles); runtime?
Ans: 7.36 s
Sol: CPI=9.2; cycles=18.4B; time=18.4/2.5

5. 1B instructions, 25% memory ops with 40-cycle stall; what % of total cycles are stalls?
Ans: 90.9%
Sol: (10B/11B)

6. In prev ques, if memory stall reduces from 40→10 cycles, what is speedup?
Ans: 3.14x
Sol: (11B / 3.5B)

7. L1=2ns (0.8 hit), L2=12ns (0.75 hit on miss), DRAM=100ns; AMAT?
Ans: 9.4ns
Sol: L1 miss rate = 0.2, L2 miss rate = 0.25, AMAT = 9.4ns

8. Cache line = 64B; how many float64 and float32 fit?
Ans: 8 and 16
Sol: (64/8;64/4)

9. 10GB CSV, only 80MB useful column, query time 25s; effective throughput?
Ans: 3.2 MB/s
Sol: (80/25)

10. In prev q, if SSD bandwidth is 500MB/s, what % bandwidth is useful?
Ans: 0.64%
Sol: (3.2/500)

11. Jobs (5,20,10) all arrive at 0; avg turnaround under FCFS vs SJF?
Ans: 21.67 vs 18.33
Sol: FCFS: Job 0 finishes at 5, job 1 finishes at 25, job 2 finishes as 35
SJF: Job 0 finishes at 5, Job 2 finishes at 15 and job 1 finishes at 35

12. Runtime: 24 min (1 worker), 5 min (6 workers); speedup and efficiency?
Ans: 4.8x, 80%
Sol: (4.8/6)

13. Total work=120, critical path=36, 4 workers; minimum completion time?
Ans: 36
Sol: max(work/4, critical path)

14. Job spends 18s compute, 27s waiting on communication; CPU utilization?
Ans: 40%
Sol: (18/(18+27))

15. Page size = 4KB; pages needed for 1MB array?
Ans: 256
Sol: (1024/4)

Longer Questions

1. A processor runs at 4 GHz. A program has:

- 50% arithmetic instructions, 1 cycle each
- 20% control instructions, 3 cycles each
- 30% memory instructions, 2 base cycles each
- Each memory instruction also stalls for 40 cycles
- Total instructions = (1.5 \times 10^9)

Questions

1. What is the average CPI without stalls?
2. What is the average CPI including stalls?
3. What is the total execution time?
4. If cache reduces stall from 40 cycles to 10 cycles, what is the speedup?

Answer

1. (0.5(1)+0.2(3)+0.3(2)=1.7)

2. (1.7+0.3(40)=13.7)
3. Cycles (=1.5B \times 13.7=20.55B); time (=20.55/4=5.14s)
4. New CPI (=1.7+0.3(10)=4.7); speedup (=13.7/4.7=2.91x)

2. In a RISC-like system:

- Each instruction is 8 bytes
- Instructions must be 8-byte aligned
- PC starts at 0x4000
- Instruction memory size is 2 KB

Questions

1. What are the next 5 instruction addresses?
2. How many instructions fit in 2 KB?
3. What fraction of all 32-bit byte addresses are valid instruction addresses?

Answer

1. 0x4000, 0x4008, 0x4010, 0x4018, 0x4020
2. (2048/8=256) instructions
3. (1/8)

3. You are working with an 8-bit signed two's complement system.

Question
Compute: (95+60)

1. Does overflow occur?
2. What 8-bit result does the machine produce?
3. What decimal value does that bit pattern represent?

Answer

1. Yes
2. (155_{10}=10011011_2)
3. (10011011_2 = -101)

Brief: max positive is 127, so 155 overflows.

4. A memory address is given as:

0x3ACF

Questions

1. Convert it to binary.
2. How many bytes long is this address?
3. If each address points to 1 byte, what is the maximum addressable memory?

Answer

1. (0011101011001111)
2. 2 bytes
3. (2^{16}=65536) bytes = 64 KB

5. A system has:

● Cache access time = 8 ns
● DRAM access time = 120 ns
● Cache hit rate = 0.75

Questions

1. What is the AMAT?
2. If hit rate improves to 0.9, what is the new AMAT?
3. What is the speedup?

Answer

1. (0.75(8)+0.25(120)=36ns)
2. (8+0.1(120)=20ns)
3. (36/20=1.9x)

6. A system has:

● L1 access = 4 ns, hit rate = 0.8
● L2 access = 20 ns, hit rate = 0.75 given L1 miss
● DRAM access = 200 ns

Questions

1. Compute the overall AMAT.
2. What fraction of accesses go to DRAM?
3. If L1 hit rate improves to 0.9, recompute AMAT.

Answer

1. 4+ 0.2(20+0.25(200)) = 18ns
2. (0.2 * 0.25 = 0.05 = 5%)
3. (4+0.1[20+0.25(200)] = 11ns)

7. Assume:

- Cache line size = 64 bytes
- Each float64 = 8 bytes
- Matrix stored in row-major order

Consider:

```
for (i = 0; i < N; i++)
  for (j = 0; j < N; j++)
    sum += A[i][j];
```

Questions

1. How many float64 elements fit per cache line?
2. Approximate fraction of accesses that are cache misses?
3. What changes if the loop order is column-wise?

Answer

1. (64/8=8)
2. About (1/8) misses, (7/8) hits
3. Column-wise causes many more misses due to strided access

8. A program executes 2 billion memory accesses.

- Cache hit time = 4 cycles
- Miss penalty = 80 cycles
- Hit rate = 0.85
- CPU clock = 2 GHz

Questions

1. Compute average cycles per memory access.
2. Compute total execution time.
3. If hit rate improves to 0.95, what is the new execution time?
4. What is the speedup?

Answer

1. (4+0.15(80)=16) cycles
2. (2B \times 16 / 2GHz = 16s)
3. New avg (=4+0.05(80)=8); time (=8s)
4. (16/8=2x)

9. Three jobs arrive on a single CPU:

Job	Arrival	Burst
P1	0	7
P2	2	4
P3	4	1

Questions

Simulate FCFS and compute average turnaround time.

Simulate non-preemptive SJF and compute average turnaround time.

Simulate SRTF and compute average turnaround time.

Answer

FCFS:

Run in arrival order: P1 → P2 → P3

0--------7----11--12

P1 P2 P3

- P1: 7 - 0 = 7
- P2: 11 - 2 = 9
- P3: 12 - 4 = 8

Avg TAT = (7+9+8)/3 = 24/3 = 8

SJF: Non-Preemptive SJF

At t=7, both P2 & P3 waiting → pick shortest (P3, burst=1)

0--------7-8--------12

- P1: 7 - 0 = 7
- P2: 12 - 2 = 10
- P3: 8 - 4 = 4

Avg TAT = (7+10+4)/3 = 21/3 = 7

SRTF: At t=7, both P2 & P3 waiting → pick shortest (P3, burst=1)

0--------7-8--------12

P1 P3 P2

- P1: 7 - 0 = 7
- P2: 12 - 2 = 10
- P3: 8 - 4 = 4

Avg TAT = (7+10+4)/3 = 21/3 = 7

10. Four jobs arrive at (t=0):

Job	Burst
P1	9
P2	5
P3	3
P4	7

Round Robin quantum = 3.

Questions

1. What is the completion time of each process?
2. What is the average turnaround time?
3. What is the average response time?

Answer
Schedule: P1 0–3, P2 3–6, P3 6–9, P4 9–12, P1 12–15, P2 15–17, P4 17–20, P1 20–23, P4

23–24
Completion: P1=23, P2=17, P3=9, P4=24
Avg TAT (=(23+17+9+24)/4=18.25)
Avg response (=(0+3+6+9)/4=4.5)

11. Assume:

- 24-bit virtual address space
- Page size = 4 KB
- Each page table entry = 4 bytes
- 8 concurrent processes

Questions

1. How many virtual pages per process?
2. How large is one page table?
3. How much memory is used for all page tables?

Answer

1. (2^{24}/2^{12}=2^{12}=4096) pages
2. (4096 \times 4 = 16384) bytes = 16 KB
3. (8 \times 16KB=128KB)

12. Assume:

- 8-bit virtual address
- Page size = 16 bytes
- VPN → PFN mapping: 0→5, 1→2, 2→7, 3→1

Questions

1. Translate virtual address 35 to physical address.
2. Translate virtual address 50 to physical address.

Answer

1. 35: VPN (=2), offset (=3), PFN (=7), physical (=7×16+3=115)
2. 50: VPN (=3), offset (=2), PFN (=1), physical (=18)

13. A process accesses pages:

A B C A D B E A B C

There are 3 physical frames, initially empty.

Questions

1. How many page faults occur under FIFO?
2. How many page faults occur under LRU?
3. Which policy does better here?

Answer

1. FIFO faults = 8
2. LRU faults = 8
3. Tie

Brief: both replace pages similarly for this short trace.

14. A 12 GB row-oriented file has 120 equal-size columns.
A query only needs one column and takes 30 seconds.

Questions

1. How much useful data is needed?
2. What is the effective useful throughput?
3. If SSD bandwidth is 600 MB/s, what percentage is useful throughput?

Answer

1. (12GB/120 = 0.1GB = 100MB)
2. (100/30=3.33MB/s)
3. (3.33/600=0.56%)

15. A task graph has total work = 100 units and critical path = 40 units.
You have 4 identical workers.

Questions

1. What is the best possible lower bound on completion time?
2. What is the maximum possible speedup?
3. If a schedule completes in 50 units, what is total idle time?

Answer

1. (\max(100/4,40)=40)
2. (100/40=2.5x)
3. (4×50-100=100) idle units

16. A program has:

- 25% serial work
- 75% perfectly parallel work
- Run on 8 cores

Questions

1. What is the maximum theoretical speedup with infinite cores?
2. What is the speedup on 8 cores?
3. Is the speedup linear?

Answer

1. (1/0.25=4x)
2. (1/(0.25+0.75/8)=2.91x)
3. No, sublinear

More Questions

1. In an 8-bit two's complement system, what is the decimal value of the bit pattern 11111110?
   a. 254
   b. -2
   c. -1
   d. 126

   Answer: Invert all bits → 00000001, add 1 → 00000010 = 2, so the original is -2.

2. A CPU cache fetches data from memory in fixed-size blocks called _____. A typical size for this unit is 64 bytes. (Ans. Cache lines. The cache loads memory in fixed 64-byte chunks to exploit spatial locality.)

3. What hardware mechanism does the OS use to reclaim the CPU from a running user process without the process's cooperation?
   a. A system call issued by the OS
   b. A hardware timer interrupt
   c. Direct memory inspection by the kernel

d. The process voluntarily yielding via a yield() call

Answer: A hardware timer interrupt A periodic interrupt fires unconditionally, forcing the kernel to regain control.

4. True or False
   a. BF16 (Brain Float 16) has a wider representable range than FP16 because it allocates more bits to the exponent field than FP16 does. (Ans. true)

      Answer: True BF16 has 8 exponent bits vs FP16's 5, matching FP32's range at the cost of precision.

5. Convert the decimal number 185 to its hexadecimal representation.
   a. What are the quotient and remainder when 185 is divided by 16? Q = 11, R = 9
   b. What is the hex digit corresponding to the quotient value from (a)? B
   c. Write the full 2-digit hex representation. 0xB9

   Answer : 0xB9 185 divided by 16 gives quotient 11 (B) remainder 9, giving B9.

6. A data pipeline has 25% of its code that is strictly sequential and 75% that achieves perfect linear speedup. The single-core runtime is 80 minutes.
   a. What is the runtime in minutes on 5 cores? 32min
   b. What is the speedup factor? 2.5x
   c. What is the theoretical maximum speedup as cores approach infinity? 4x

   Answer: 32 min, 2.5x, 4x T = (0.25 + 0.75/5) x 80 = 32 min; Amdahl max = 1/0.25 = 4x.

7. Four processes arrive at time 0 with burst times: P1 = 12 ms, P2 = 3 ms, P3 = 8 ms, P4 = 5 ms. Using Shortest Job First (non-preemptive) scheduling, what is the average turnaround time?
   a. 14.0 ms
   b. 17.5 ms
   c. 22.0 ms
   d. 13.75 ms

   Answer: SJF order: P2(3), P4(5), P3(8), P1(12). Turnaround times: P2=3, P4=3+5=8, P3=3+5+8=16, P1=3+5+8+12=28. Average = (3+8+16+28)/4 = 55/4 = 13.75 ms.

8. A system uses 32-bit virtual addresses and a page size of 4 KB.
   a. How many bits are used for the page offset? 12 bits
   b. How many bits are used for the virtual page number? 20 bits
   c. How many entries are in a flat single-level page table for one process? 1048576 entries

   Answer: 12 bits, 20 bits, 1,048,576 entries

   4 KB = 4096 = 2^12, so 12 offset bits. 32 - 12 = 20 VPN bits. 2^20 = 1,048,576 entries

9. A programmer writes two loops to sum a 1000x1000 integer matrix stored in row-major order in memory: Version A: for i in rows: for j in cols: total += M[i][j] Version B: for j in cols: for i in rows: total += M[i][j]
   a. Which version runs faster on a modern CPU? Explain why in terms of cache lines and spatial locality. A

b. What cache behavior does the slower version exhibit? Cache Thrashing

Answer: Version A, cache thrashing

Row-major storage means M[i][0], M[i][1]... are contiguous. Version A walks sequentially through memory. Version B accesses M[0][j], M[1][j]... which are 1000 integers apart, causing a cache miss every access. "Cache thrashing" is the right term here.

10. Consider a custom 6-bit float: Bit 5 = sign (s), Bits 4-3 = exponent (2 bits), Bits 2-0 = fraction (f2, f1, f0).Value = (-1)^s * 2^(exp - 1) * (1 + f2*2^-1 + f1*2^-2 + f0*2^-3)What is the decimal value of the bit sequence 010110?
    a. 1.75
    b. -1.75
    c. 3.5
    d. -3.5

    Answer: 3.5 s=0 (positive), exp=10 in binary=2 so 2^(2-1)=2, fraction 110 gives 1+0.5+0.25=1.75, and 2 x 1.75 = 3.5.

11. A dataset has 500 million rows with schema (UserID int64, Timestamp int64, Value float64). Row-store format, no compression. Use 1 GB = 10^9 bytes.
    a. What is the size of one row in bytes? 24 bytes
    b. What is the total file size in GB? 12GB
    c. A batch gradient descent run reads the full file 10 times (10 epochs) across 3 different learning rates. What is the total read I/O in GB? 360GB

    Answer: 24 bytes, 12 GB, 360 GB

    int64 + int64 + float64 = 8+8+8 = 24 bytes. 500M x 24 = 12,000,000,000 bytes = 12 GB. 12 x 10 x 3 = 360 GB.

12. A program stores temperatures in an 8-bit signed integer (two's complement).
    a. What is the maximum positive value representable? 127
    b. The current reading is 120. A delta of +15 is added. What decimal value is actually stored? -121
    c. Name the phenomenon that occurred and describe one real-world consequence in a safety-critical system. Signed Integer Overflow

    Answer: **127, -121, signed integer overflow**

    Max 8-bit signed = 2^7 - 1 = 127. 120 + 15 = 135. 135 in binary = 10000111. Interpreted as signed = -128 + 7 = -121.

13. Dataset D: n = 200 million rows, d = 4 numeric features plus a label Y, all stored as float64 in row-store format. 1 GB = 10^9 bytes.

    Step A: Build a linear model using Batch Gradient Descent. Try 3 learning rates, 4 epochs each.

    Step B: Read D, append all quadratic pairwise feature interactions (no squared terms), write expanded file to disk.

    Step C: Rebuild using same model selection as Step A but reading the expanded file.

    a. How many new feature columns does Step B add? 6
    b. What is the size of file D in GB? 8GB

c. What is the size of the expanded file written in Step B in GB? 17.6GB
d. What is the total I/O cost in GB across all three steps (reads and writes)? 332.8GB

Answer: 6 columns, 8 GB, 17.6 GB, 332.8 GB

C(4,2) = 6. File D = 5 cols x 8 bytes x 200M = 8 GB. Expanded = 11 cols x 8 bytes x 200M = 17.6 GB. Step A reads = 8 x 4 x 3 = 96 GB. Step B = 8 read + 17.6 write = 25.6 GB. Step C reads = 17.6 x 4 x 3 = 211.2 GB. Total = 96 + 25.6 + 211.2 = 332.8 GB.

14. A single-core system runs four processes: P1=20ms, P2=5ms, P3=15ms, P4=10ms. All arrive at time 0. Context switch overhead is 1ms per switch (charged before each process except the first).
    a. Under FIFO scheduling (order P1,P2,P3,P4), what is the average waiting time? 22.75ms
    b. Under SJF (non-preemptive) scheduling, what is the average waiting time? 14ms
    c. How many milliseconds total are consumed by context switches under each algorithm? 3ms each

    Answer: **22.75 ms, 14 ms, 3 ms each**

    FIFO with context switches: P1 starts at 0 (no switch), P2 starts at 20+1=21, P3 starts at 21+5+1=27, P4 starts at 27+15+1=43. Waiting times: P1=0, P2=21, P3=27, P4=43. Average = 91/4 = 22.75 ms. SJF order P2(5), P4(10), P3(15), P1(20): P2 starts at 0, P4 at 5+1=6, P3 at 6+10+1=17, P1 at 17+15+1=33. Waiting: 0+6+17+33 = 56/4 = 14 ms. Context switches = 3 ms each.

15. A multi-threaded analytics job runs on a 2-socket NUMA server. Remote RAM access takes 3x longer than local RAM access. Local RAM latency = 100 ns. Thread A (on Socket 0) reads a 4 GB array stored in Socket 1's RAM. Thread B (on Socket 1) reads a 4 GB array stored in Socket 1's RAM.
    a. What effective latency does Thread A experience per memory access? 300ns
    b. Both threads write to adjacent 64-bit integers that share the same 64-byte cache line. What performance problem arises? Name this phenomenon. False Sharing
    c. A data engineer proposes copying Thread A's 4 GB array to Socket 0's RAM before the job starts. What is the trade-off? Faster local access but 4 GB extra RAM + one-time copy cost

    Answer: 300 ns, false sharing, trade-off

    3 x 100 = 300 ns. False sharing is correct since both threads share a cache line.

QUESTION=module, COUNT=10, CATEGORY=

For the following questions, choose one option that best represents the answer. Note that an incorrect choice carries negative marks.

QUESTION=singlecorrect, MARKS=1.00

Pipelining as a strategy is used to reduce the latency of jobs in the system. True or False?

OPTIONS=

1. False
2. True

ANSWER=A

QUESTION=singlecorrect, MARKS=1.00

Which of the following results in a high hit rate for an on-core cache?

OPTIONS=

1. Size of Cache
2. Size of the DRAM
3. cache access speed
4. DRAM Access speed

ANSWER=A

ANSDESC=

QUESTION=singlecorrect, MARKS=1.00

A process P1 will transition from BLOCKED to READY (RUNNABLE) under which of the following events?

OPTIONS=

1. When the I/O issued by P1 completes
2. When P1 issues a call to do I/O
3. When a new process P2 arrives in the system
4. When P1 arrives for the first time (when P1 is born).

ANSWER=A

QUESTION=singlecorrect, MARKS=1.00

Two processes on a system will NOT share which of the following?

OPTIONS=

1. Any part of the logical address space
2. The physical memory (DRAM)
3. the CPU
4. the Disk

ANSWER=A

QUESTION=singlecorrect, MARKS=1.00

CPU Scheduling is done so as to:

OPTIONS=

1. Decrease CPU utilization and have more spare CPU around
2. Allow multi-programming and take advantage of the IDLE time of the CPU
3. Increase the speed of a process and therefore improve its turnaround time
4. Increase utilization of the DRAM on the computer.

ANSWER=B

QUESTION=singlecorrect, MARKS=1.00

Having a low response time is particularly important for

OPTIONS=

1. Batch jobs (those that are submitted, take a long time to run and return with all results at once)
2. Interactive jobs
3. Neither batch nor interactive jobs
4. Both batch AND interactive jobs

ANSWER=B

QUESTION=singlecorrect, MARKS=1.00

Increasing the page size will result in which of the following?

OPTIONS=

1. increasing the external fragmentation and lowering internal fragmentation
2. increasing the internal fragmentation while not affecting the external fragmentation
3. increasing the internal fragmentation while lowering the external fragmentation
4. increasing both the internal and external fragmentation

ANSWER=B

QUESTION=singlecorrect, MARKS=1.00

When *thrashing* occurs, it

OPTIONS=

1. increases the CPU utilization

2. decreases the CPU utilization
3. does not impact CPU utlization but it does increase the size of the logical address space
4. does not affect either the CPU utlization or the size of the address space

ANSWER=B

ANSDESC=

QUESTION=singlecorrect, MARKS=1.00

The 2s complement of decimal 12 is:

OPTIONS=

1. 11110100
2. 0100
3. 01000
4. 00000100

ANSWER=A

ANSDESC=

12 is 1100 and can be represented in 4 bits. However for negative 12 (Which the twos complement represents) cannot be. So we must use at least 5 bits. The algorithm to derive 2s complement is to invert all bits and add 1.

QUESTION=numeric, MARKS=1.00

If we devised a new floating point representation in 16 bits (instead of the 32 we studied in class) and assigned 4 bits to the exponent, the biasing factor to be applied will be _____ (recall the biasing factor we used in a 32 bit FP standard was 127 - do you recall why?)

ANSWER=[7.0, 7.0]

ANSDESC=

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QUESTION=module, COUNT=2, CATEGORY=

Assume a 5 stage perfectly balanced instruction execution pipeline with a 10ms (10 millisecond) stage time.

QUESTION=numeric, MARKS=2.00

How many milliseconds does it take to complete the **first** ten instructions?

ANSWER=[140.0, 140.0]

ANSDESC=

The first instruction finishes in 50 milliseconds. Thereafter an instruction completes every 10 ms. Hence 10 instructions complete in 140 ms.

QUESTION=numeric, MARKS=3.00

Now assume that the 5th instruction is a branch that was mispredicted and the error is realized at the end of the third stage of the pipeline. What this means is that the assumption of a branch being taken or not is incorrect and found out when that instruction completes stage 3. At this point the pipeline has to be flushed and whatever is in the pipeline for instructions 6 through 10 must be executed from the beginning (i.e., whatever was in the pipeline is flushed out and the pipeline starts afresh). Given this mess, how long will take to complete 10 instructions? Assume instruction number 5 (which was the branch) does not have to be reexecuted - it continues its merry way and completes.

ANSWER=[160.0, 160.0]

ANSDESC=

So, the first instruction completes in 50ms as before. Instructions 2-5 will complete at 60, 70, 80ms and 90ms respectively. Now, the 6th instruction which was in stage 2 when instruction 5 was in stage 3 and the 7th which was in stage 1 have to be flushed out and new instructions 6-10 start entering the pipeline at the 70 ms mark. So the 6th instruction exits the pipeline at 120 ms and 7 through 10 exit at 130, 140, 150 and 160 ms respectively. So it now takes 160 ms instead of the earlier 140.

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QUESTION=paper, MARKS=6.00

Consider the following processes, arrival times, and CPU processing requirements:

PID	Arrival Time	Processing Time (Amount of CPU Time)
1	0	3

2	1	5
3	3	2
4	9	2

For each of the following scheduling algorithms, fill in the table with the process that is running on the CPU (for timeslice-based algorithms, assume a 1 unit timeslice): For RR, assume that an arriving thread is scheduled to run at the beginning of its arrival time. I have filled in the first row. The empty rows indicate the PID of the process scheduled by each algorithm in a specific time unit. Each cell is worth 0.25 points except for the turnaround time which is worth 0.5 each for a total of 6 points. **Upload an image of the completed table as your answer.**

Time	**Round Robin**	**SRCTF**
0	1	1
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
Average Turnaround time		

ANSDESC=

Time, RR, SRCTF

0 1 1

1 2 1

2 1 1

3 3 3

4 2 3

5 1 2

6 3 2

7 2 2

8 2 2

9 4 2

10 2 4

11 4 4

Average turnaround times

RR = 5.75

SRCTF = 4.25

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QUESTION=numeric, MARKS=4.00

This question is about physical addressability in a system that uses paging. Assume the following:

- Its a 20 bit byte addressable computer.
- Page size is 4KB.
- The system we're running upon has 1 GB of physical memory (a GB is 2^{32} bytes).
- For efficiency, the page table is organized as an array of as many entries as there are logical pages per process. Each array entry is a bitstring that captures the physical frame

number as well as 12 additional bits for bookkeeping purposes (such as dirty, validty etc.)

How much space in memory does the OS use for page tables assuming that 16 concurrent processes are executing. **Express your answer in terms of number of physical frames needed. If needed, round your answer to the next highest integer value.**

Hint: Work in powers of two.

ANSWER=[4.0, 4.0]

ANSDESC=

with a 20 bit computer, we have a logical address space of 2^20 bytes. IF each page can hold 2^12 bytes then we have a maximum of 2^8 pages per process in the logical space and so 2^8 entries per process in the page table. Each entry needs to have a physical frame number which takes up 20 bits and 12 more bits for the bookkeeping information as per the information given. Hence each entry is 32 bits (2^2 bytes). Hence the total space for one process needed for the page table is 2^8 * 2^2/2^12 pages. If we have 16 processes which is 2^4 processes we have a total of 4 pages used up.

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QUESTION=module, COUNT=2, CATEGORY=

This question pertains to demand paging and page replacement algorithms. Assuming that the access string is a repeated one such as ABCDABCDABCD where each reference is to a page and the number of pages after which it repeats varies. The example is for a 4 page access.

QUESTION=numeric, MARKS=3.00

If the number of pages accessed is 4 (ABCD repeated.....) and the number of Physical frames available is 3, the probability of a page fault if LRU were used is _____. Assume the reference string is infinite which means the probabilities converge.

ANSWER=[1.0, 1.0]

ANSDESC=

Will work it out in class.

QUESTION=numeric, MARKS=3.00

If we had 3 frames and the same access string (ABCD repeated for ever) but we employed the OPT policy instead, the probability of a page fault converges to_____

ANSWER=[0.3, 0.4]

CLASS PROBLEMS

A processor runs at 3 GHz. A program has the following instruction mix:

- 40% arithmetic instructions, 1 cycle each
- 30% memory instructions, 5 cycles each
- 30% control instructions, 2 cycles each

Total instructions executed = 2×10^9
Questions
1. What is the average CPI (Clock cycles per instruction)?
2. What is the total number of cycles?
3. What is the execution time?
4. If memory instructions increase to 50%, with arithmetic dropping to 20% and control staying at 30%, what is the new execution time?

In a ARM-like system:
- Each instruction is 4 bytes
- Instructions must be 4-byte aligned

PC starts at 0x1000
Questions
1. What are the next 5 instruction addresses?
2. How many instructions fit in 1 KB of instruction memory?
3. What fraction of all 32-bit addresses are valid instruction addresses?

A program executes:
- 1 billion instructions
- 30% are memory operations
- Each memory operation stalls for 100 cycles
- CPU clock = 2.5 GHz
- Non-memory instructions take 1 cycle
- Assume memory instructions also take 1 base cycle plus the stall

Questions
1. How many cycles are spent on base instruction execution?
2. How many cycles are spent waiting on memory?
3. What is the total execution time?
4. What percentage of total cycles are memory stall cycles?
5. If cache reduces the stall from 100 cycles to 20 cycles, what is the speedup?

You are working with an 8-bit signed two’s complement system.

Question
Compute: 120+10
1. Does overflow occur?
2. What result does the machine actually produce?

A memory address is given as:
0x7F3A
Questions
1. Convert it to binary
2. How many bytes does this address represent?
3. If each address points to 1 byte, what is the maximum addressable memory?

Two level hierarchy: CPU cache and Memory (DRAM) only:
CPU cache access time: 10 ns
Memory access time: 1.5 microsecond
CPU Cache hit rate: 0.7
What is the Average Memory access time (AMAT)?
How about for a three-level hierarchy: L1 cache on core (10ns), L2 cache on CPU(70ns) and DRAM(1.5 microsec)?

A system has:
- Cache access time = 10 ns
- Main memory access time = 1.5 μs
- Cache hit rate = 0.7

Questions
1. What is the average memory access time (AMAT)?
2. If hit rate improves to 0.9, what is the new AMAT?
3. What is the speedup?

A system has:
- L1 access = 10 ns, hit rate = 0.7

- L2 access = 70 ns, hit rate = 0.9 (given L1 miss)
- DRAM access = 1.5 μs

Questions

1. Compute the overall AMAT
2. What fraction of accesses go to DRAM?
3. If L1 hit rate improves to 0.8, recompute AMAT

Assume:

- Cache line size = 64 bytes
- Each integer = 4 bytes
- Array stored in row-major order

Consider:

```
for (i = 0; i < N; i++)
    for (j = 0; j < N; j++)
        sum += A[i][j];
```

Questions

1. How many elements are loaded per cache line?
2. What fraction of accesses are cache hits (approximate)?

Now consider column-wise access:

```
for (j = 0; j < N; j++)
    for (i = 0; i < N; i++)
        sum += A[i][j];
```

Estimate how cache misses change.

A program executes 1 billion memory accesses.

- Cache hit time = 5 cycles
- Memory access penalty (on miss) = 100 cycles
- Hit rate = 0.8
- CPU clock = 2 GHz

Questions

1. Compute average cycles per memory access
2. Compute total execution time
3. If hit rate improves to 0.95, what is the new execution time?
4. What is the speedup?

Whats the output of the following program?

```
// W(A) means print the letter A to the console
int main() {
	W(A);
	int child = fork();
	W(B);
	if(child)
		wait(NULL);
	W(C);
}
```

Scenario: You are the OS of a deep-space dispatch station. Six ships have arrived and need CPU time to complete their mission computations. Simulate FCFS, Non Preemptive SJF, SRTF and RR scheduling algorithms and compare the results.

Ship	**Arrival Time**	**Burst Time**	**Job Type**
Orion	0	8	Cargo routing
Vega	1	4	Emergency beacon
Pulsar	2	9	Deep scan
Nova	3	5	Medical supply
Rift	4	2	Fuel request
Echo	6	6	Crew transfer

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4 Cores, 12 jobs, all arrive at t=0ms. RR Quantum is 4 ms.

Jobs

Job	Burst	Priority
J1	20 ms	High
J2	4 ms	Low
J3	12 ms	Low
J4	8 ms	High
J5	16 ms	Low
J6	6 ms	High
J7	10 ms	Low
J8	2 ms	High
J9	14 ms	Low
J10	18 ms	High
J11	4 ms	Low
J12	8 ms	High

Part A — SQMS (single global queue)

Q1	How many total context switches before all jobs complete?
Q2	What is the makespan (time until last job finishes)?
Q3	J1 has burst 20 ms. How many times is it preempted?
Q4	What is the average waiting time across all 12 jobs?
Q5	J13 (burst 6 ms) arrives at t = 8 ms. When does it first run?

Part B — MQMS (one queue per core)

Core 0	Core 1
J1 · J5 · J9 Total: 50 ms	J2 · J6 · J10 Total: 28 ms
Core 2	**Core 3**
J3 · J7 · J11 Total: 26 ms	J4 · J8 · J12 Total: 18 ms

Q1	What is the makespan for each core? Which is the bottleneck?
Q2	How long is each non-bottleneck core idle at the end?
Q3	Load imbalance ratio = longest / shortest queue. Compute it.

Q: A data science pipeline reads a 2 GB CSV file from a spinning hard disk drive with the following characteristics:

•Rotational speed: 7200 RPM

•Average seek time: 8 ms

•Transfer rate: 150 MB/s

•Block size: 4 KB

The file is stored in two ways in separate scenarios:

Scenario A: The file is fully contiguous on disk.

Scenario B: The file is fragmented into 512 equal-sized chunks scattered randomly across the disk.

Answer the following:

1.Calculate the average rotational latency in milliseconds.

2.Calculate the total I/O time for **Scenario A**.

3.Calculate the total I/O time for **Scenario B**. *(Assume each chunk requires a fresh seek and rotational delay.)*

4.How many times slower is Scenario B than Scenario A? What does this imply for how data scientists should think about file formats and storage layout?